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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
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Branch	CSE
Course Outcome	Co-3

Case Study Topic: Retweet Reach Prediction in X (Twitter) using BFS with Visited Set Optimization

Work Done Summary:

- Implemented BFS on the Twitter follower graph starting from source user A with depth limit 3.
- Maintained a visited set to avoid double-counting users reachable through multiple paths.
- Processed neighbors in alphabetical order for deterministic traversal.
- Recorded queue evolution, discovered neighbors, and visited-set updates at each iteration.
- Identified multi in-edge nodes and verified that visited set prevents duplicate counting.
- Computed final reach set and analyzed BFS efficiency for bounded-depth search.

Implementation Details:

- Represented the follower network as a directed graph using adjacency list.
- Used a queue for BFS traversal and a visited set to track discovered users.
- Explored neighbors in alphabetical order at each level.
- Enforced depth limit of 3 to simulate 10-minute retweet reach prediction.
- For each neighbor: if not visited, added to queue and marked visited.
- Captured queue evolution and visited set after every dequeue operation.
- Identified predecessors for nodes with multiple in-edges.
- Computed final set of reached users (excluding source).
- Measured reach count and analyzed overlapping audiences handling.

Result: Screenshots / Output

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BFS traversal from A up to depth 3
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Step 1: Dequeued A (depth 0)    -> discovered: [B, C]    Visited: {A, B, C}
Step 2: Dequeued B (depth 1)    -> discovered: [D, E]    Visited: {A, B, C, D, E}
Step 3: Dequeued C (depth 1)    -> discovered: [E, F]    (E already visited) Visited: {A, B, C, D, E, F}
Step 4: Dequeued D (depth 2)    -> discovered: [G]      Visited: {A, B, C, D, F, G}
Step 5: Dequeued E (depth 2)    -> discovered: [G, H]    (G already visited) Visited: {A, B, C, D, E, F, G, H}
Step 6: Dequeued F (depth 2)    -> discovered: [H, I]    (H already visited) Visited: {A, B, C, D, E, F, G, H, I}
Step 7: Dequeued G (depth 3)    -> depth limit reached
Step 8: Dequeued H (depth 3)    -> depth limit reached
Step 9: Dequeued I (depth 3)    -> depth limit reached
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Final reached users (excluding A): {B, C, D, E, F, G, H, I}
Total unique reach: 8
Multiple In-Edge Nodes and Predecessors:
E <- B, C    (Chosen predecessor: B)
G <- D, E    (Chosen predecessor: D)
H <- E, F    (Chosen predecessor: E)
I <- F       (Chosen predecessor: F)
```

The BFS traversal correctly reached 8 unique users within depth 3. The visited set prevented double-counting for E, G, H, and I which are reachable through multiple paths.

Top 5 recent timestamps returned:

50400, 46800, 43200, 39600, 36000

Time Complexity:

- BFS (AVL Insert/Delete): $O(\log n)$

- Top K Query: $O(\log n + k)$

This approach provides accurate bounded-depth reach prediction with optimized performance for large-scale social graphs.

GitHub Link:

<https://github.com/pondurutanvitha-stack/DSA-2>

Student Signature	Faculty Signature